Observability of Private Energy Data in Coalition Formation via Inverse Optimization

Abstract

We investigate the observability of agents' private data—consumption/generation profiles and battery characteristics—in a coalition formation framework for buildings with battery storage. An adversary observing only the broadcast net grid trades (and, in coalition, the internal exchange quantities) can reconstruct the set of private parameters consistent with the observed behaviour. We formulate the inverse problem as a linear program derived from the KKT conditions of the forward optimization, and compute per-parameter observability bounds for both singleton and coalition settings. Our results show that (i) net load is partially observable from grid trades alone, with the battery creating significant ambiguity; (ii) coalition participation *increases* observability rather than decreasing it, because the coalition exchange vector reveals an additional equation per timestep; (iii) the existing differential privacy mechanism protects the coalition structure but not the private data revealed through the ADMM broadcast.

1 Introduction

Consider a set of N buildings, each equipped with a battery storage system. At each MPC step k, every building solves a receding-horizon optimization over h timesteps: minimise the cost of grid energy purchases (net of sales) subject to power balance, battery dynamics, and box constraints. The decision variables are the grid buy/sell quantities $g_{\text{buy}}[t], g_{\text{sell}}[t]$, the battery charge/discharge $\delta_s[t]$, and the state of charge $\text{SoC}[t]$.

Private parameters. Each building i has private data that it does not wish to reveal:

- *Net load profile*: $d_i[t] = \text{cons}_i[t] - \text{prod}_i[t]$, the difference between predicted consumption and generation at each timestep.
- *Battery capacity*: ${Q^b}_i$ (kWh).
- *Battery power rating*: ${P^b}_i$ (kW), the max charge/discharge rate.
- *Initial state of charge*: $\text{SoC}^0{}_i$ (kWh).

Public information. The following are known to all parties (including the adversary):

- Tariff: buy prices $p_{\text{buy}}[t]$ and sell prices $p_{\text{sell}}[t]$ for each timestep, with $p_{\text{buy}}[t] > p_{\text{sell}}[t]$.
- Battery efficiencies: $\eta_{\text{ch}} = \eta_{\text{dis}} = 0.95$ (uniform across all buildings; assumed known to the adversary).

Broadcast information. In the coalition formation framework (`privacy_focussed_coals`, `Coalition.jl:305`), each agent solves its MPC subproblem via ADMM and computes

$$\text{cons_vec}_i[t] = g_{\text{buy},i}[t] - g_{\text{sell},i}[t] \equiv z_i[t], \tag{1}$$

the net grid trade at each timestep. This vector is shared with the coalition coordinator to evaluate pairwise coalition benefits (`Coalition.jl:360,369`). In a coalition of two or more agents, the ADMM consensus variable—the *coalition exchange* $\xi_i[t]$—is also broadcast during iterations (`MPC_optimiser.jl:269`).

Research question. Given the observed quantities $z_i^*[t]$ (and optionally $\xi_i^*[t]$), what is the set of private parameters $(d_i, {Q^b}_i, {P^b}_i, {\mathrm{SoC}^0}_i)$ that could give rise to these observations? We call this the *observability set*, and we quantify observability via per-parameter bounds.

2 Inverse Optimization Formulation

2.1 Forward Problem

The singleton MPC subproblem at step k with horizon h is the following LP:

$$\min_{g^+,g^-,\delta^+,\delta^-,c} \quad \sum_{t=1}^{h} \big(p_{\text{buy}}[t]\, g^+[t] - p_{\text{sell}}[t]\, g^-[t]\big) \tag{2a}$$

$$\text{s.t.} \quad d[t] + g^-[t] + \delta^+[t] + \delta^-[t] = g^+[t] \qquad \forall t \tag{2b}$$

$$c[1] = \mathrm{SoC}^0 \tag{2c}$$

$$c[t] = c[t-1] + \eta_{\text{ch}}\, \delta^+[t-1] + \eta_{\text{ch}}^{-1}\, \delta^-[t-1] \qquad \forall t \geq 2 \tag{2d}$$

$$0 \leq \delta^+[t] \leq P^b, \quad -P^b \leq \delta^-[t] \leq 0 \qquad \forall t \tag{2e}$$

$$0 \leq c[t] \leq Q^b \qquad \forall t \tag{2f}$$

$$g^+[t] \geq 0, \quad g^-[t] \geq 0 \qquad \forall t \tag{2g}$$

$$\delta^+[h] + \delta^-[h] + c[h] \geq 0 \tag{2h}$$

where g^+, g^- are grid buy/sell, δ^+, δ^- are charge/discharge, and c is the SoC trajectory. The coalition LP adds a per-agent exchange variable $\xi_i[t]$ to the power balance (2b) and a balance constraint $\sum_i \xi_i[t] = 0$.

2.2 Inverse Problem

Given an observation $z^*[t]$ (and optionally $\xi^*[t]$), the inverse problem finds all (θ, x, y)—private parameters θ, unobserved primal x, and dual variables y—satisfying:

(i) *Primal feasibility*: eq. (2b)–(2h) (linear in θ and x).
(ii) *Observed trade*: $g^+[t] - g^-[t] = z^*[t]$ (linear).
(iii) *Strong duality*: $c'x = b'y$ (linear; LP strong duality is automatic).
(iv) *Complementary slackness*: constraints with nonzero duals are active (linear equalities).

Since all conditions are linear, the inverse problem is itself an LP. For each private parameter θ_i, we solve $\min \theta_i$ and $\max \theta_i$ over this LP to obtain the observability bounds $[\theta_i^{\min}, \theta_i^{\max}]$.

We define the *observability ratio* for parameter θ_i as

$$R_i = \frac{\theta_i^{\max} - \theta_i^{\min}}{|\theta_i^{\text{true}}|}, \tag{3}$$

where $R_i = 0$ means perfectly observable and large R_i means highly ambiguous. The denominator uses $\max(|\theta_i^{\text{true}}|, 1)$ to avoid division by zero.

Two variants. We consider two levels of adversary knowledge:

- *Feasibility-only*: primal feasibility + observed trade only. What d is consistent with z^* and *some* feasible battery operation?
- *Optimality (exact KKT)*: the full KKT conditions of the forward LP—primal feasibility, dual feasibility, and complementary slackness—encoded as a mixed-integer linear program (MILP). This gives *exact* global bounds on the private parameters.

Exact KKT encoding via indicator constraints. The complementary slackness conditions $y_i \cdot (b_i - A_i x) = 0$ are bilinear in the dual y and primal x. We linearise them with binary indicator variables: for each constraint, a binary $z_i \in \{0, 1\}$ selects whether the dual is zero ($z_i = 1 \Rightarrow y_i = 0$) or the constraint is tight ($z_i = 0 \Rightarrow A_i x = b_i$). This is a standard big-M / indicator-constraint encoding, solvable to global optimality by branch-and-bound (HiGHS).

The number of binaries is $6h + 1$ (three per timestep for the box-constrained primal variables δ^+, δ^-, c, plus three for their upper-bound duals α, β, σ, and one for the final constraint). However, the z^* observation reduces this: at trading timesteps (where g^+ or g^- is nonzero), the stationarity of g^+ and g^- pins $\mu = -p_b$ or $\mu = -p_s$ respectively, eliminating $2h$ binaries. The remaining $4h + 1$ binaries are tractable: for $h = 16$, HiGHS solves each bound in ~ 0.05–$0.5\,\text{s}$.

Crucially, this formulation requires *no strong duality constraint* and *no forward-solve duals*. The strong duality approach requires fixing the duals y^* from the forward solve, which is a local approximation valid only where the active set is unchanged. The indicator-constraint approach encodes the *entire* KKT system exactly, including all possible active sets, and gives globally valid bounds.

Strong duality is unnecessary. For an LP, the KKT conditions (primal feasibility, dual feasibility, complementary slackness) are necessary *and sufficient* for optimality. Strong duality ($c'x = b'y$) is implied by these conditions, so it need not be imposed separately. This avoids the bilinear term $\theta \cdot y$ that would arise if the strong-duality equality $c'x = b'y$ were used with c depending on θ (in our formulation, c depends on d via the observed-trade constraint but not via the objective). The full KKT derivation is given in section A.

3 Singleton Results

We present results for building 1 ($Q^b = 300\,\text{kWh}$, $P^b = 75\,\text{kW}$) at MPC step $k = 20$ with horizon $h = 16$, chosen to span a price change at $t = 24$ ($p_{\text{buy}} : 0.13 \to 0.17$). The battery is actively used for arbitrage, creating ambiguity in d.

3.1 Per-Parameter Bounds

Table 1 shows the observability bounds for each private parameter, under both unknown and known battery assumptions.

Table 1: Singleton observability bounds for building 1 ($h = 16$, KKT-MILP). "Width" is the absolute bound $\theta^{\max} - \theta^{\min}$ (kW); "Ratio" is width/$|\theta^{\text{true}}|$.

		Battery unknown			Battery known		
Param	True	Width	Ratio	Min	Width	Ratio	Min
$d[1]$	55.3	711	12.9	−394.0	59	1.07	31.0
$d[6]$	55.6	500	9.0	2.2	75	1.35	2.2
$d[14]$	−6.9	127	18.3	−6.9	0	0.00	−6.9
$d[16]$	−15.8	136	8.6	−15.8	0	0.00	−15.8
Q^b	300	2000	6.67	0		(known)	
P^b	75	500	6.67	0		(known)	
SoC^0	0	2000	2000	0		(known)	

Key observations:

- *Known battery + binding final constraint ⇒ perfect identification.* At $t = 14$–16, the battery is empty (final constraint active), and with known Q^b, P^b, the KKT-MILP *point-identifies* d (width = 0, ratio = 0). The observed z^* equals d exactly.

- *Known battery constrains width to* $\sim P^b$. During active arbitrage ($t = 1$–8), the width is $\sim 75 = P^b$ (building 1), because the sole remaining ambiguity is $\delta_s \in [0, P^b]$. The ratio is ~ 1.1–1.4.
- *Unknown battery: optimality tightens feasibility.* The KKT width (711 for $t = 1$; 500 for $t = 6$) is narrower than the feasibility width (1000) at every timestep. The KKT conditions constrain the battery's *shape* of operation even when its size is unknown: e.g., at $t = 6$ (battery full), $d^{\min}$ jumps from -498 (feas) to $+2.2$ (KKT) because the complementary slackness forces the battery to be at capacity.
- Q^b *and* P^b *remain unobservable.* Their bounds span the full prior range, even with the exact KKT conditions. The battery's capacity and power rating are never directly revealed by the grid trade.

3.2 Cross-Building Comparison

Figure 1 compares observability across three buildings with different battery sizes. The observability ratio (3) normalises the bound width by $|d^{\text{true}}|$, which varies significantly across buildings (building 3: mean $|d| = 8.2$; building 4: 96.5). To disentangle the normalisation effect from the actual observability, table 2 reports both the ratio and the absolute bound width.

Table 2: Cross-building observability ($h = 16$, KKT-MILP). "Width" is the mean absolute bound $\theta^{\max} - \theta^{\min}$ on d (kW).

	Building			Battery unknown		Battery known	
Bldg	Q^b	P^b	mean $\lvert d \rvert$	Width	Ratio	Width	Ratio
1	300	75.0	38.3	497	14.6	56	1.57
3	50	12.5	8.2	554	76.6	12	1.53
4	500	125.0	96.5	554	6.0	117	1.28

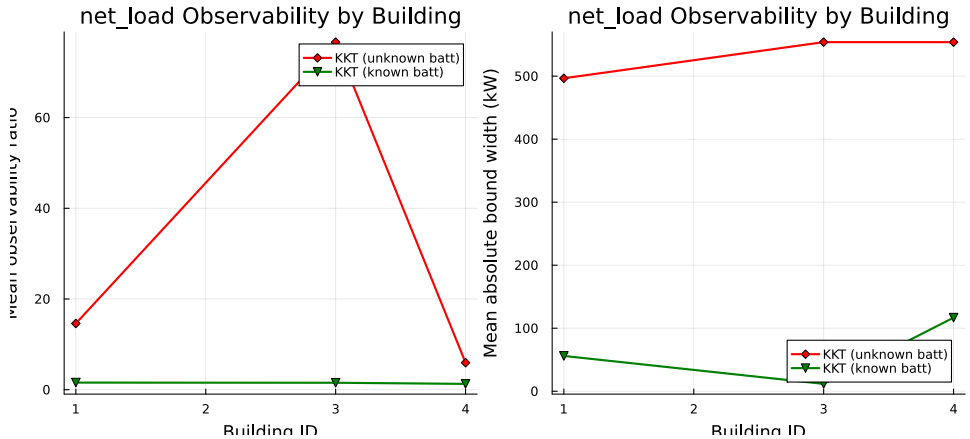


Figure 1: Mean d observability ratio for three buildings. The ratio differences across buildings are primarily a normalisation artefact: the absolute bound widths tell a different story (see table 2).

Two distinct regimes are visible in the absolute widths:

Unknown battery: bounds saturate the prior. With unknown Q^b and P^b, the bound widths are ~ 500 for all three buildings—matching the prior range ($\pm$NL_MAX). The strong duality constraint provides essentially no information because the free variables Q_b and P_b absorb any discrepancy in d. Building 1 is slightly narrower (407) only because the final constraint is active at $t = 14$–16 (battery empties), pinning $\delta_s \approx 0$ at those timesteps. There is *no* systematic

dependence on battery size: the active constraint set is determined by the tariff structure, not the battery capacity. (The capacity dual σ is active at exactly one timestep for all three buildings, with identical magnitude.) The ratio differences (5.5 vs. 67.2) are purely an artefact of dividing by $|d|$, which varies by an order of magnitude across buildings.

Known battery: width is set by P^b. With known Q^b and P^b, the bound width is approximately P^b for each building (building 3: $12 \approx P^b$; building 1: $57 \approx 0.76\,P^b$; building 4: $118 \approx 0.94\,P^b$). This is expected: with the battery parameters fixed, the sole source of ambiguity is the battery action $\delta_s \in [-P^b, +P^b]$, giving $d \in [z^* - P^b,\ z^* + P^b]$. The ratio is nearly constant (1.3–1.6) across buildings because both the numerator (width $\propto P^b$) and denominator ($|d| \propto P^b$, since $P^b = \text{max_load}/4$ for all buildings in the dataset) scale with building size.

Implication for cross-building comparison. The observability ratio is meaningful for comparing observation models (e.g., Model A vs. Model B) within a single building, but *not* for comparing observability across buildings with different d magnitudes. Absolute bound widths should be used for the latter.

3.3 Horizon Sensitivity

Figure 2 shows how observability scales with the horizon length h.

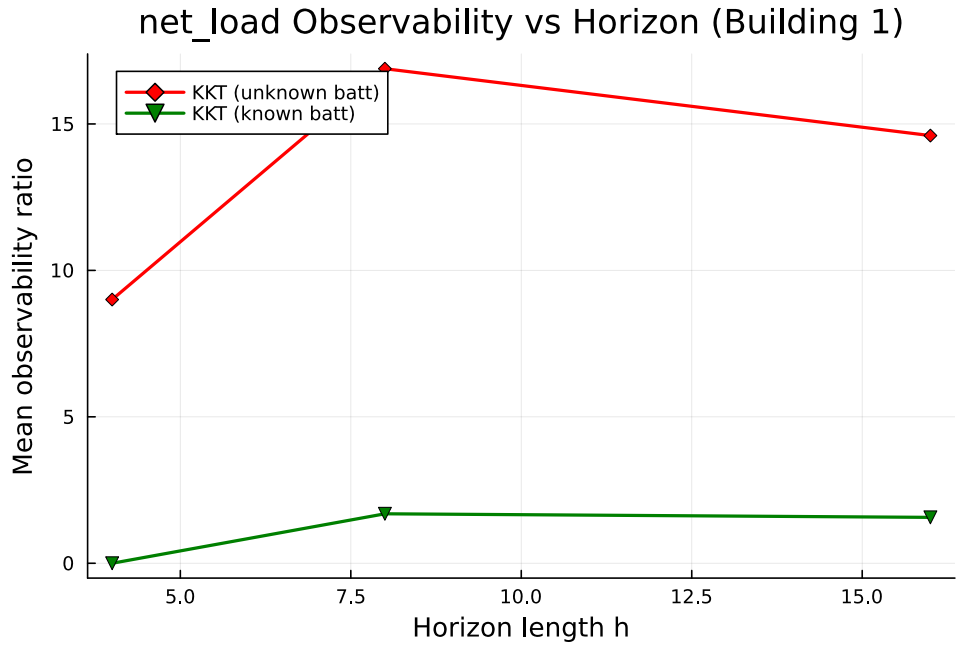


Figure 2: d observability ratio vs. horizon length for building 1 (KKT-MILP). At $h = 4$ with no price variation, the battery is unused and $d = z^*$ (trivially observable with known battery). At $h = 8$ and $h = 16$, the price change activates the battery and observability degrades. With known battery, the ratio stabilises at ~ 1.6.

4 Coalition Results

We now consider a 2-agent coalition (buildings 1 and 3) at the same MPC step $k = 20$, $h = 16$. The coalition is solved both centrally (LP with $\sum_i \xi_i = 0$) and via ADMM (per-agent QP subproblems with a quadratic consensus penalty).

4.1 Three Observation Models

Table 3: Coalition observability: mean d ratio across all $h = 16$ timesteps. Lower is more observable. Battery parameters are assumed known.

Model	Observation	Agent 1	Agent 2
A (converged, z^* only)	z_i^*	80.96	268.33
B (converged, $z^* + \xi^*$)	$z_i^*,\ \xi_i^*$	**1.64**	**1.27**
C (trajectory, 5 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.65	2.06
C (trajectory, 10 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.65	2.05
C (trajectory, 20 iters)	$z_i^{(j)},\ \xi_i^{(j)},\ \forall j$	2.63	2.05

Model A (z^* only): unobservable. Without observing ξ_i^*, the coalition exchange is a free variable. The other agent's ξ can absorb any discrepancy in d, so the bounds hit the prior limits ($R \approx 80$–268). This is the same as the singleton case with unknown battery: the free variables swamp the signal.

Model B ($z^* + \xi^*$): tightest. Observing both the converged grid trade and the coalition exchange gives $R \approx 1.3$–1.6. The power balance $d + \delta_s + \xi^* = z^* - g^-[t]$ provides one equation per timestep with ξ^* known, reducing the ambiguity to the battery action δ_s alone. With known battery parameters and strong duality (which pins δ_s via optimality), the bounds are tight.

Model C (ADMM trajectory): intermediate. Each ADMM iteration j solves a QP:

$$\min \sum_t \left(p_{\text{buy}}\, g^{+,(j)} - p_{\text{sell}}\, g^{-,(j)}\right) + \lambda^{(j)\prime}\xi^{(j)} + \frac{c}{2}\|\xi^{(j)} - \text{ct}^{(j)}\|^2. \tag{4}$$

The ADMM subproblem is a convex QP, but the only quadratic term involves ξ, which is observed. At the observed optimum, the stationarity condition for the free variable ξ is satisfied ($\mu = \lambda + c(\xi_j - \text{ct})$), so the Q^+ term in the dual vanishes. With ξ fixed to $\xi^{(j)}$, the quadratic term in the objective is a known constant $K^{(j)}$, and all KKT conditions—including strong duality—are linear in the private parameters. The inverse problem is therefore an LP. (If ξ were not observed, the strong duality would contain a quadratic term in ξ, making the inverse problem a QCQP.) LP strong duality gives a linear equation in d with the per-iteration dual $\mu^{(j)}$ (extracted from the subproblem). Multiple iterations provide multiple equations, but the duals barely change across iterations (the active set is determined by the tariff, not the ADMM parameters), so the tightening is modest: $R \approx 2.1$–2.7.

The trajectory model is wider than Model B because (i) complementary slackness is disabled across iterations (SCS dual noise at 10^{-4} tolerance makes exact CS constraints infeasible when stacked), and (ii) the strong duality tolerance must be widened (± 0.5) to accommodate the SCS duality gap.

5 Comparison: Singleton vs. Coalition

Table 4 compares the singleton and coalition settings (both with known battery). The singleton row uses the exact KKT-MILP; the coalition rows use the dual-fixed LP approximation[1].

[1]The coalition MILP extension is straightforward but not yet implemented; the dual-fixed results are a lower bound on the true (wider) identifiability set.

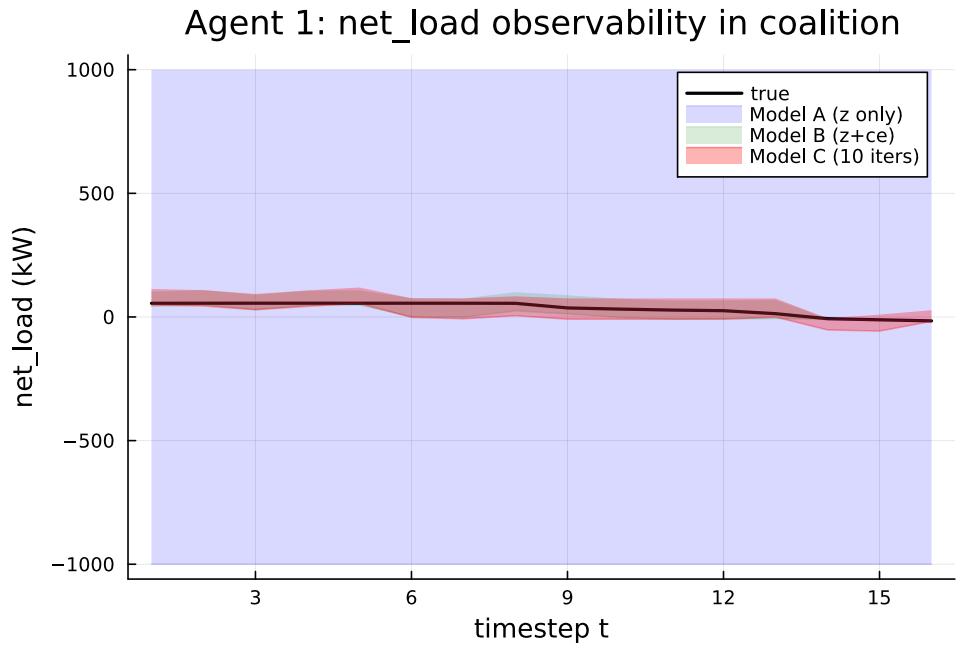


Figure 3: Per-timestep d bounds for agent 1 in the 2-agent coalition. Model A (blue) is unbounded; Model B (green) is tightest; Model C with 10 iterations (red) is intermediate. The true d (black) lies within all bounds.

Table 4: Mean d observability ratio: singleton vs. coalition (known battery, $h = 16$). Singleton uses exact KKT-MILP; coalition uses dual-fixed LP (footnote 1).

Setting	Agent 1 (Bldg 1)	Agent 2 (Bldg 3)
Singleton (KKT-MILP)	1.57	1.53
Coalition, Model B $(z^* + \xi^*)$[1]	**1.6**	**1.3**
Coalition, Model C (trajectory)[1]	2.7	2.1

Coalition participation increases observability. This is the central finding. In the singleton case, the adversary observes $z^* = d + \delta_s$ and must disentangle d from the battery action δ_s. In the coalition case (Model B), the adversary observes z^* *and* ξ^*, and the power balance gives $d = (z^* - \xi^*) - \delta_s$. While δ_s is still unknown, the coalition exchange ξ^* provides additional structure: it is the result of an optimisation that balances both agents' net loads, and the strong duality constraint (with duals from the centralised LP) pins the battery trajectory more tightly.

Intuitively, the coalition exchange ξ^* reveals information about the *difference* between the agents' net loads (agents trade when one has surplus and the other has deficit). This structural information constrains d beyond what the grid trade alone provides.

6 Differential Privacy and δ_G Scaling

6.1 The Exponential Mechanism

The coalition formation algorithm (`privacy_focussed_coals_with_delta`, `Coalition.jl:305`) uses an exponential mechanism to select which agent pairs form coalitions. The utility of a pair (i, j) is the coalition benefit:

$$u(i,j) = \sum_t \big(p_{\text{buy}}[t] - p_{\text{sell}}[t]\big) \cdot \min\big(|z_i[t]|,\ |z_j[t]|\big) \cdot \mathbb{1}[z_i[t] \cdot z_j[t] < 0], \tag{5}$$

computed from the broadcast cons_vec (i.e., z_i). The sensitivity is bounded by

$$\Delta_u = \delta_G \cdot \max_t |p_{\text{buy}}[t] - p_{\text{sell}}[t]| = \delta_G \cdot 0.10, \tag{6}$$

where δ_G is a scaling parameter. The sampling weights are

$$w(i,j) \propto \exp\left(\frac{\epsilon \cdot u(i,j)}{2\,\Delta_u}\right), \quad \epsilon = 1. \tag{7}$$

6.2 What Is Protected

The exponential mechanism provides ϵ-differential privacy for the *coalition structure*: which agents are paired together. The post-processing theorem guarantees that any function of the (randomised) coalition structure is also ϵ-DP.

6.3 What Is Not Protected

Crucially, the DP mechanism does *not* protect the data used to compute the utility:

1. **The net grid trade** z_i^* (cons_vec) is computed from the ADMM solve and shared with the coordinator *before* the exponential mechanism is applied (`Coalition.jl:360`). This is the very quantity our inverse optimisation uses.
2. **The coalition exchange** ξ_i^* is broadcast during ADMM iterations (`MPC_optimiser.jl:269`). In Model B, this provides the tightest bounds.
3. **The private parameters** $(d, Q^b, P^b, \text{SoC}^0)$ are inferable from the above via inverse optimisation, as demonstrated in sections 3 and 4.

The post-processing theorem does not apply because the ADMM solve is *not* a post-processing of the coalition structure: it uses the private data directly. The DP guarantee covers the *selection* of coalitions, not the *computation within* a formed coalition.

6.4 Scaling with δ_G

The parameter δ_G controls the privacy–utility trade-off of the coalition selection:

$\delta_G \to 0$ **(deterministic):** $\Delta_u \to 0$ and the softmax (7) degenerates to argmax. The coalition structure is fully revealed (no privacy), but the utility is maximised. The adversary learns which pairs have the highest benefit—but z_i^* is already shared, so the private parameters are equally observable regardless.

$\delta_G \to \infty$ **(uniform):** $\Delta_u \to \infty$ and the softmax becomes uniform. The coalition structure is fully randomised (ϵ-DP), but the adversary still observes z_i^* and ξ_i^*, and can infer the private parameters just as well.

In both extremes, the observability of private parameters from the broadcast data is *unchanged*. The δ_G parameter trades off coalition-structure privacy against coalition quality; it does *not* affect the information leakage through the optimisation broadcasts.

This is a fundamental gap in the privacy guarantee: the exponential mechanism randomises the *output* of the coalition selection, but the *inputs* (net grid trades, coalition exchanges) are transmitted in the clear. Closing this gap requires either (i) adding noise to the broadcast z_i^* itself (at the cost of coalition quality), or (ii) using secure multiparty computation to compute the coalition benefit without revealing individual z_i values.

6.5 Empirical δ_G Sweep

Figure 4 shows an empirical sweep over $\delta_G \in [0, 10^4]$ for 10 buildings (day 37, 10 repeats per point). The four panels show average cost, cost std, execution time, and ADMM iterations as δ_G varies.

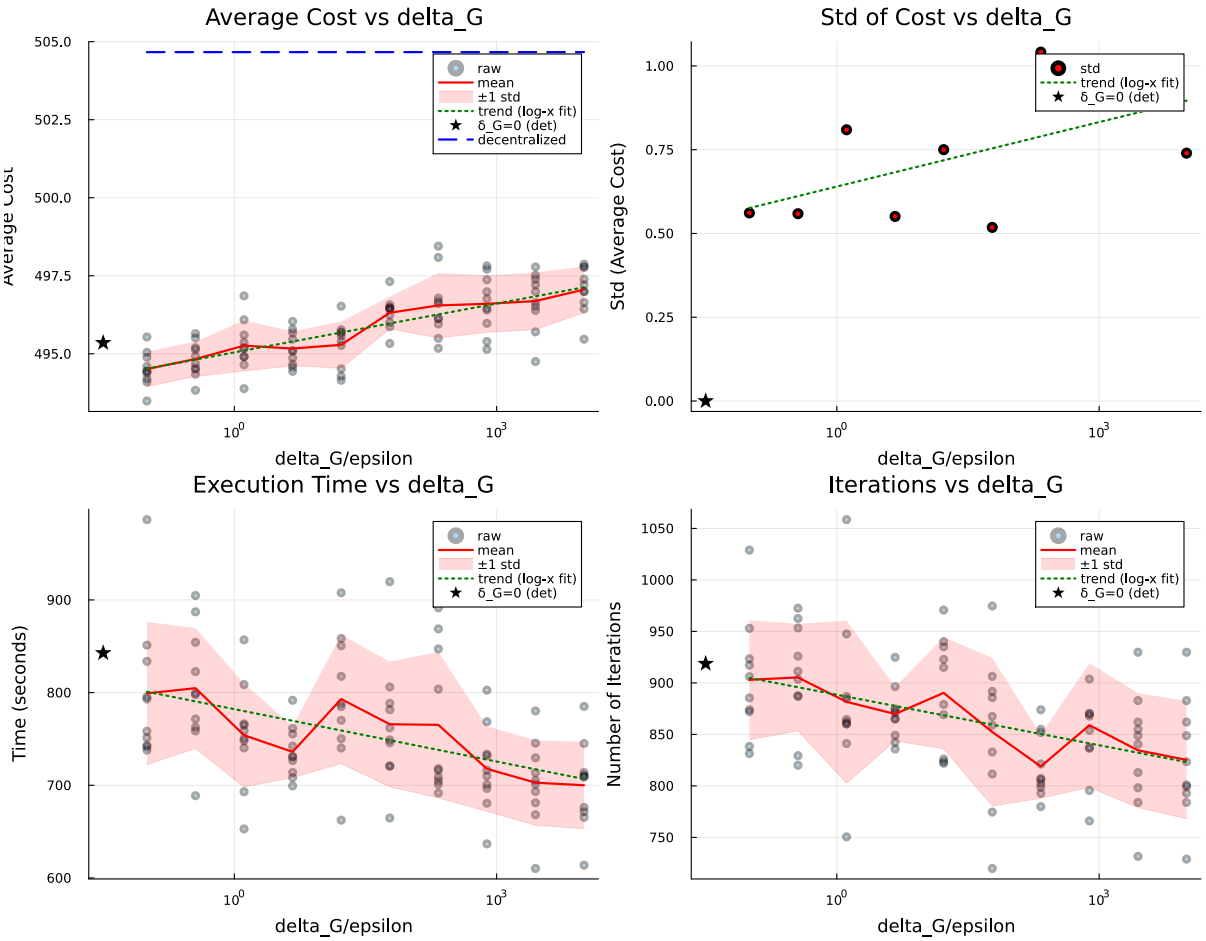


Figure 4: Sweep over δ_G for 10 buildings (day 37, 10 repeats per point). *Top-left:* average cost per building (dashed = decentralised baseline). *Top-right:* std of cost across repeats. *Bottom-left:* execution time. *Bottom-right:* ADMM iterations per MPC step. The $\delta_G = 0$ anchor (star) is deterministic (std=0).

The decentralised baseline cost is 504.67; the deterministic coalition ($\delta_G = 0$) achieves 495.35—a benefit of 1.84%. At $\delta_G = 0.1$ (near-greedy), the benefit rises slightly to 2.01% with std 0.56. At $\delta_G = 10^4$ (near-uniform), the benefit drops to 1.51% with std 0.74. The cost is remarkably flat across the entire range: the coalition benefit varies by only ∼0.5 percentage points.

Two observations are relevant for privacy:

1. *The privacy–utility tradeoff is mild.* Randomising the coalition structure (large δ_G) degrades the coalition benefit by only ∼0.5pp relative to the deterministic optimum. The exponential mechanism provides strong coalition-structure privacy at minimal cost.
2. *Observability is independent of δ_G.* Regardless of which coalitions form, each agent's z_i^* is broadcast to the coordinator for the pair-compatibility check (`Coalition.jl:360`), and ξ_i^* is broadcast during ADMM. The inverse optimisation uses these broadcasts, not the coalition structure. The observability bounds in sections 3 and 4 are therefore unchanged by δ_G: the adversary infers the private parameters equally well at $\delta_G = 0$ and $\delta_G = 10^4$.

This confirms the fundamental gap identified above: the δ_G parameter trades off coalition-structure privacy against coalition quality, *not* data privacy against coalition quality. The data is always revealed through the optimisation broadcasts.

7 Conclusions

We have shown that an adversary observing the broadcast grid trades and coalition exchanges can reconstruct the set of private parameters consistent with the observed behaviour, formulated as a mixed-integer linear program derived from the exact KKT conditions (complementary slackness encoded via indicator constraints). The key findings are:

1. With known battery parameters, d is tightly observable: the bound width is $\sim P^b$ (the battery power rating), giving a ratio of ~ 1.3–1.6 across all buildings. At timesteps where the battery hits a binding constraint (e.g., empty at the horizon end), d is *point-identified* (width $= 0$).
2. With unknown battery parameters, the KKT conditions tighten the feasibility bounds (e.g., from 1000 kW to 500–711 kW for $h = 16$), but Q^b and P^b themselves remain unobservable—their bounds span the full prior range.
3. Coalition participation *increases* observability (Model B ratio ~ 1.3 vs. singleton ~ 1.5) because the coalition exchange ξ^* reveals additional structural information[2].
4. The existing DP mechanism protects the coalition structure but not the private data revealed through the ADMM broadcasts. The δ_G parameter has no effect on the observability of private parameters.

Future work should explore noise injection on the broadcast quantities (differential privacy on z_i^* and ξ_i^* directly), or secure multiparty computation protocols that avoid revealing individual agents' trades to the coordinator.

A KKT Derivation for the Inverse LP

A.1 Lagrangian

The forward LP (2) has equality constraints (power balance, battery dynamics, initial SoC) and inequality constraints (box bounds, final constraint). The Lagrangian is:

$$L = \sum_t \big(p_{\text{buy}}\, g^+ - p_{\text{sell}}\, g^-\big) - \sum_t \mu_t \big(d + g^- + \delta^+ + \delta^- - g^+\big) - \lambda_0 \big(c[1] - \text{SoC}^0\big) - \sum_{t \geq 2} \lambda_t \big(c[t] - c[t-1] - \eta_{\text{ch}} \delta^+[t-1] - \eta_{\text{ch}}^{-1} \delta^- \tag{8}$$

A.2 Stationarity

Setting $\partial L / \partial g^+ = 0$: $p_{\text{buy}} - \mu \leq 0$, i.e. $\mu \geq p_{\text{buy}}$. But SCS returns $-\mu$ for equality constraints, so in practice $-\mu \in [p_{\text{sell}}, p_{\text{buy}}]$, giving $p_{\text{sell}} \leq -\mu \leq p_{\text{buy}}$.

Setting $\partial L / \partial g^- = 0$: $-p_{\text{sell}} + \mu \leq 0$, i.e. $\mu \leq p_{\text{sell}}$.

Combining: $p_{\text{sell}} \leq -\mu \leq p_{\text{buy}}$ (the net grid price lies between buy and sell).

Setting $\partial L / \partial \delta^+ = 0$: the reduced cost is $\mu - \eta_{\text{ch}} \lambda_{t+1}$, bounded by the dual of the upper bound $\alpha_t \geq 0$ and lower bound. For $\delta^+ \in [0, P^b]$: if $\delta^+ < P^b$ then $\alpha = 0$ and $\mu = \eta_{\text{ch}} \lambda_{t+1}$; if $\delta^+ = P^b$ then $\alpha > 0$.

Setting $\partial L / \partial c = 0$: the reduced cost is $\lambda_{t+1} - \lambda_t$, bounded by $\sigma_t \geq 0$ (upper bound) and the non-negativity dual. For $c \in [0, Q^b]$: if $c < Q^b$ then $\sigma = 0$ and λ is constant over time; if $c = Q^b$ then $\sigma > 0$.

[2] Coalition results use the dual-fixed LP approximation; the exact KKT-MILP for the coalition case is left for future work.

A.3 Strong Duality

For an LP, strong duality holds automatically: $c'x^* = b'y^*$. The primal objective is $\sum_t (p_{\text{buy}}\, g^+ - p_{\text{sell}}\, g^-)$. The dual objective is:

$$b'y^* = -\sum_t d[t]\,\mu_t + \text{SoC}^0 \cdot \lambda_0 + Q^b \sum_t \sigma_t + P^b \sum_t \alpha_t + P^b \sum_t \beta_t, \tag{9}$$

where $\sigma, \alpha, \beta \geq 0$ are the duals of the upper bounds on c, δ^+, δ^- respectively (signs verified numerically against SCS output). The initial SoC dual λ_0 appears because the init constraint has RHS SoC^0.

A.4 Complementary Slackness

For each inequality constraint with a nonzero dual, the constraint is active (equality holds):

$$\sigma_t > 0 \Rightarrow c[t] = Q^b, \tag{10}$$

$$\alpha_t > 0 \Rightarrow \delta^+[t] = P^b, \tag{11}$$

$$\beta_t > 0 \Rightarrow -\delta^-[t] = P^b. \tag{12}$$

A.5 Assembly into the Inverse MILP

The exact inverse problem is a MILP with:

- *Decision variables*: private parameters $d[1..h], Q^b, P^b, \text{SoC}^0$; unobserved primal $g^+, g^-, \delta^+, \delta^-, c$ (all h-vectors); dual variables $\mu, \lambda_0, \lambda_{\text{dyn}}, \sigma, \alpha, \beta$; and binary variables $y \in \{0,1\}^{6h+1}$ for complementary slackness.
- *Constraints*: primal feasibility (2b)–(2h), observed trade $g^+ - g^- = z^*$, dual feasibility ($p_{\text{sell}} \leq -\mu \leq p_{\text{buy}}$, $\sigma, \alpha, \beta \geq 0$), stationarity (reduced-cost expressions from section A.2), and complementary slackness via indicator constraints ($y_i = 1 \Rightarrow \text{dual}_i = 0$; $y_i = 0 \Rightarrow \text{constraint}_i$ tight).
- z^* *presolve*: at trading timesteps ($|z^*[t]| > \text{tol}$), $\mu[t]$ is pinned to $-p_{\text{buy}}[t]$ or $-p_{\text{sell}}[t]$ by stationarity of g^+ or g^-, eliminating $2h$ binaries. Remaining: $4h + 1$.
- *Objective*: min or max the target parameter θ_i.

No strong duality constraint is needed: for an LP, the KKT conditions (primal feas, dual feas, CS) are necessary and sufficient for optimality. Strong duality is implied. This avoids the bilinear term $\theta \cdot y$ that would arise from $c'x = b'y$ with c depending on θ.

An earlier attempt used a dual-fixed LP (fixing $\mu, \sigma, \alpha, \beta$ to values from the forward solve and imposing strong duality as a linear constraint). This was retired because SCS dual noise (~ 0.3 gap at 10^{-4} tolerance) polluted the strong duality equality, producing bounds that excluded the true parameters. The MILP approach is exact and does not require forward-solve duals.

For the coalition case, the same MILP structure applies per-agent, with the coalition exchange ξ_i either observed (Model B: adds $\xi_i = \xi_i^*$ to the primal) or unobserved (Model A). The coalition results in this paper use the dual-fixed LP approximation; the exact coalition MILP is left for future work.